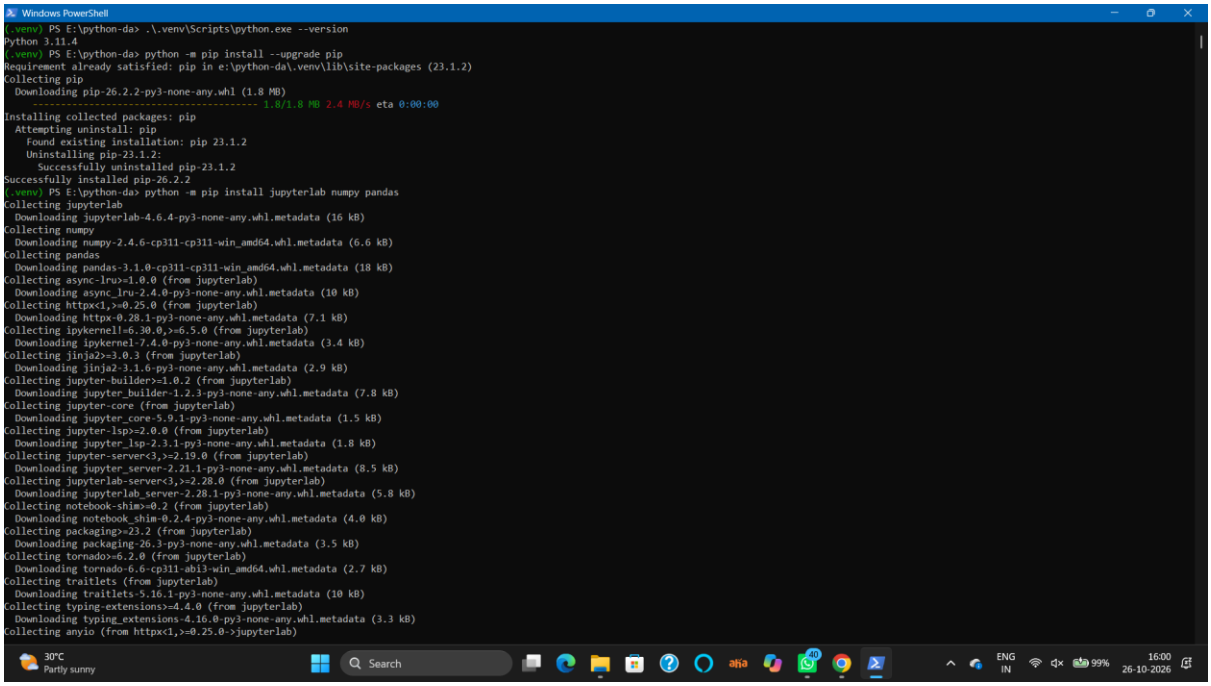
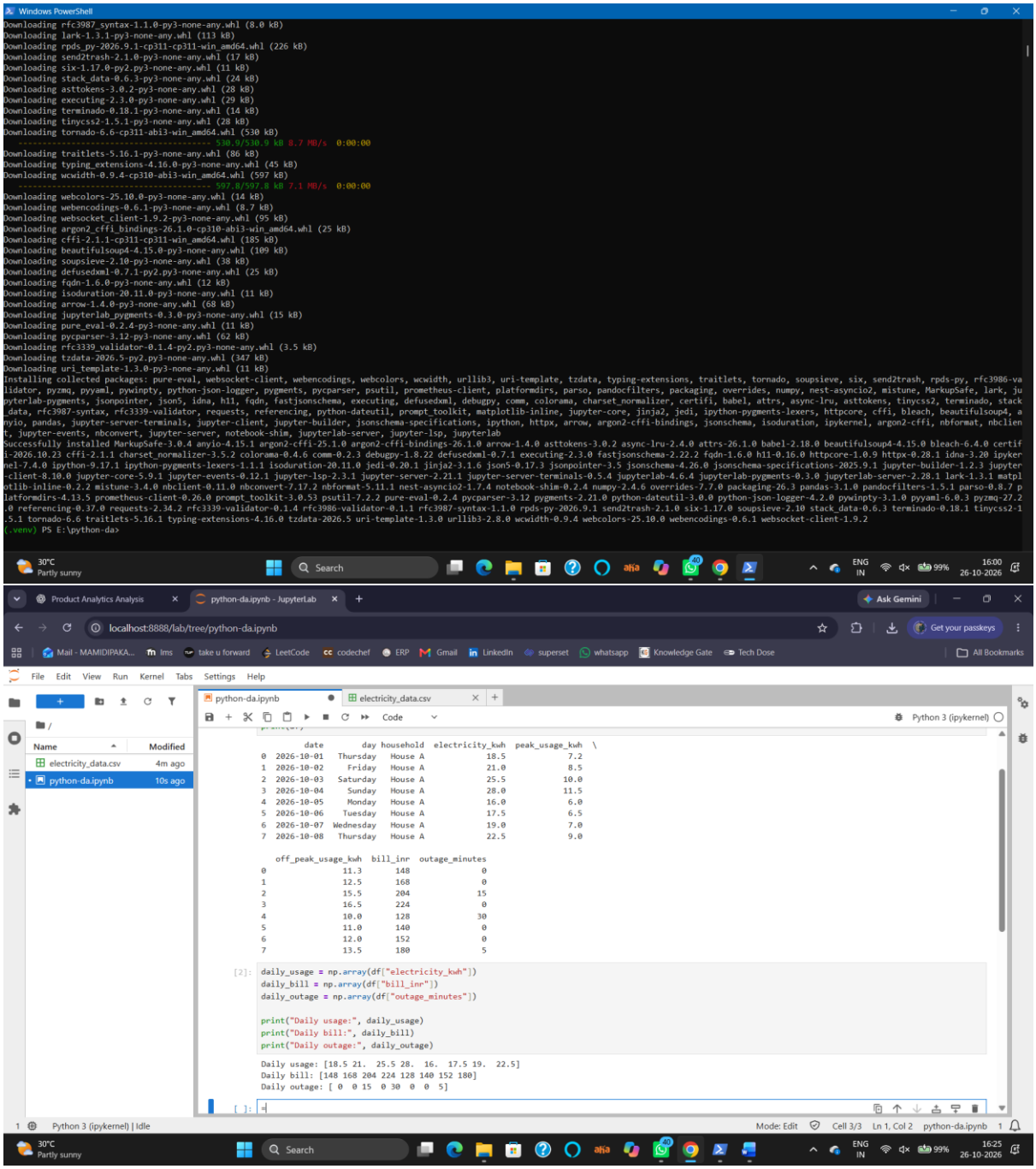
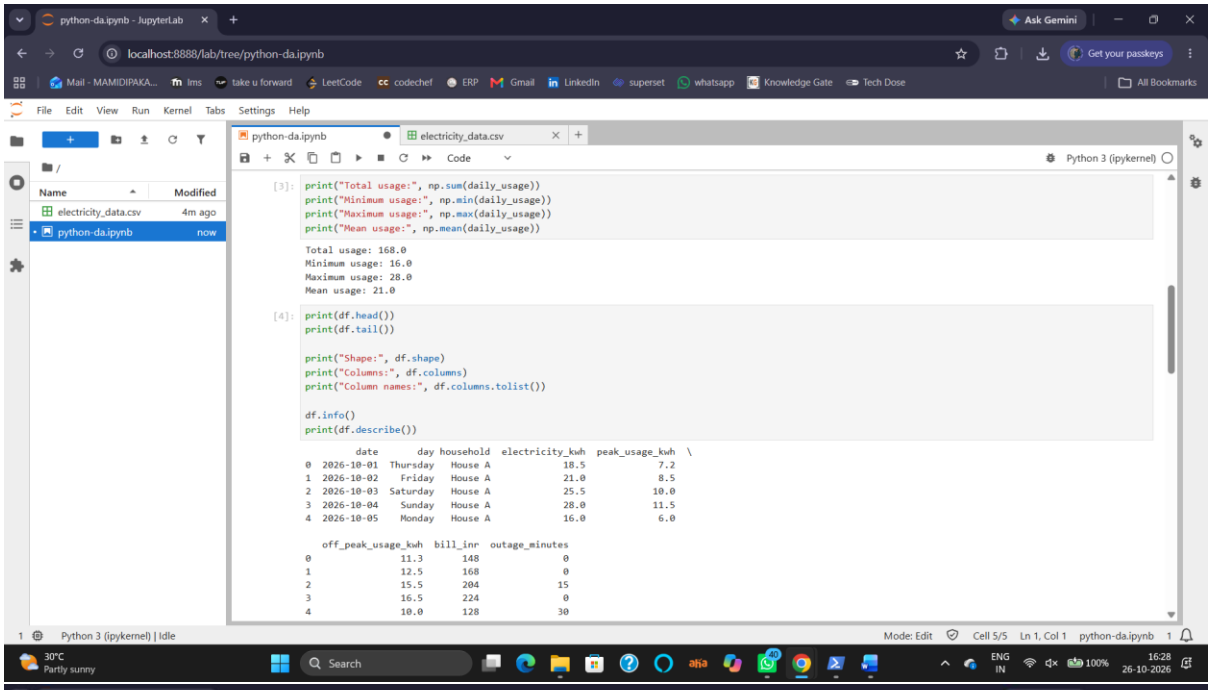
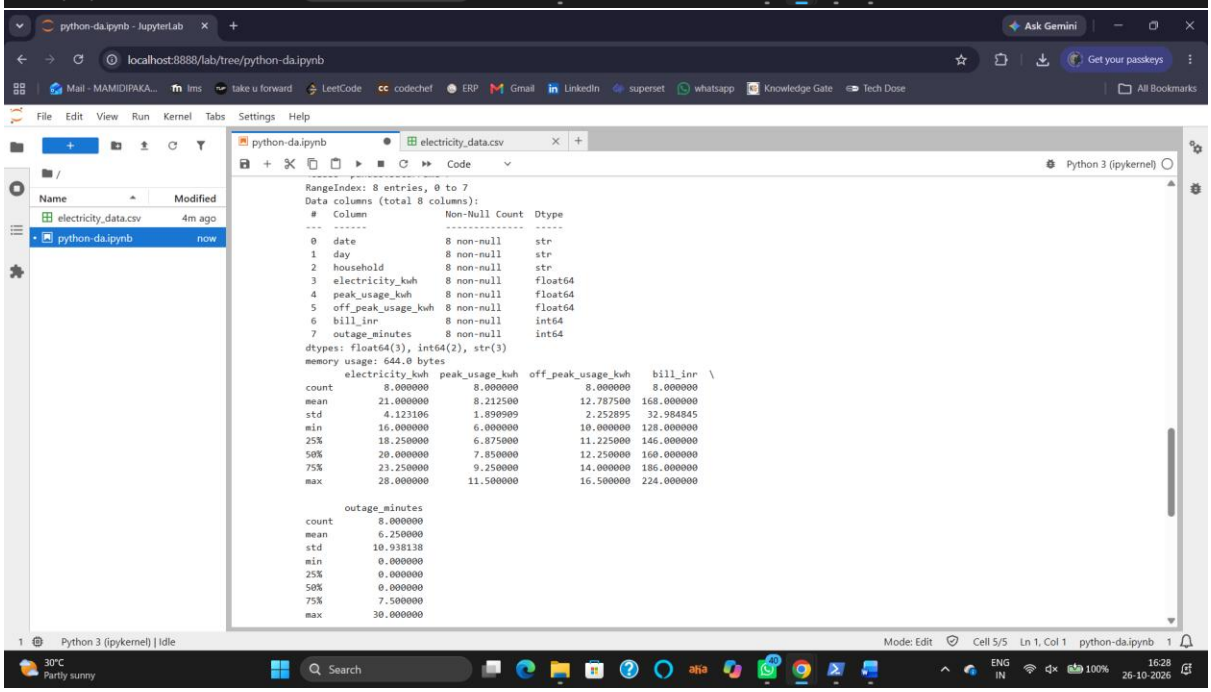
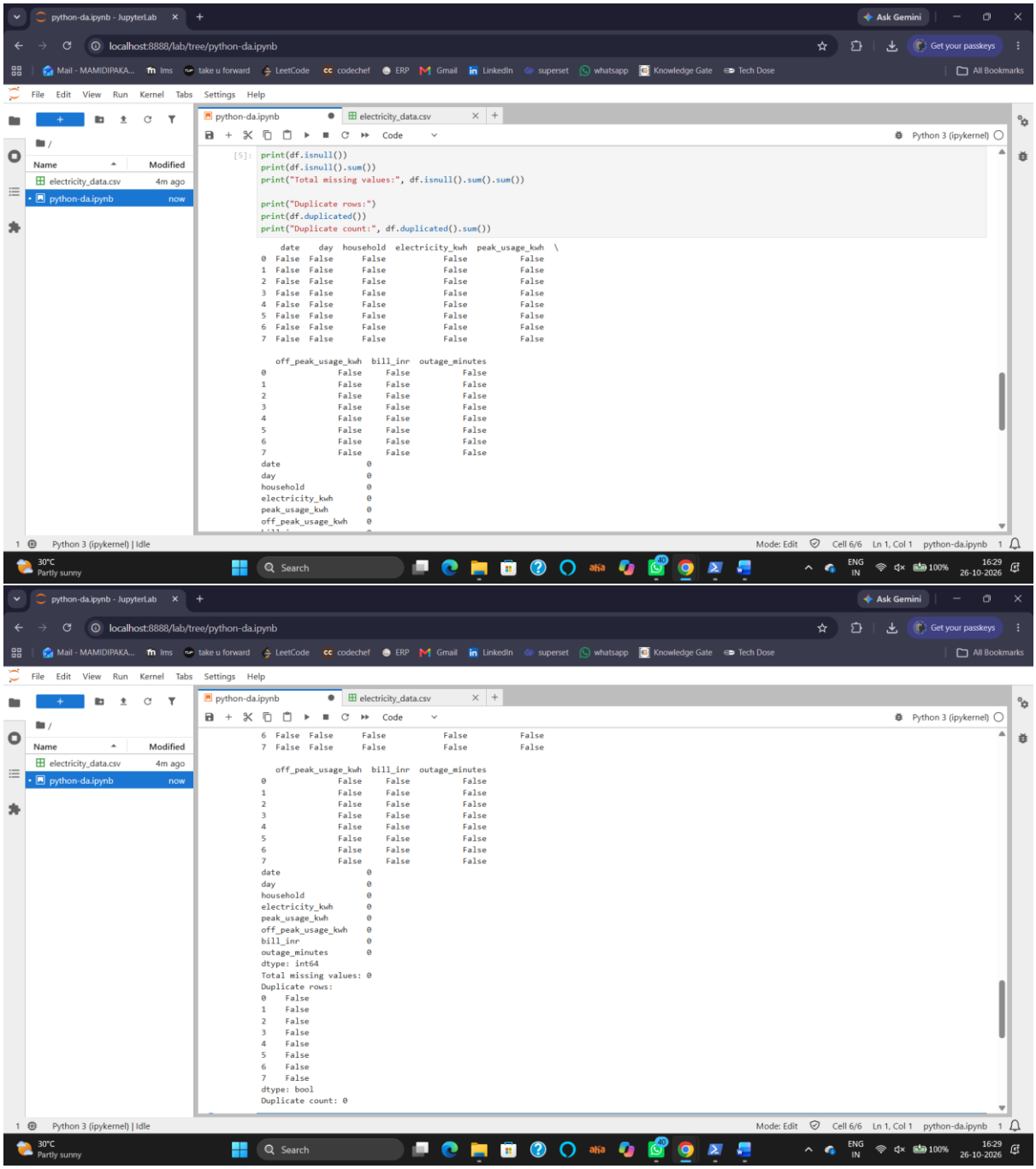

Windows PowerShell
(.venv) PS E:\python-da> .\.venv\Scripts\python.exe --version
Python 3.11.4
(.venv) PS E:\python-da> python -m pip install --upgrade pip
Requirement already satisfied: pip in e:\python-da\.venv\lib\site-packages (23.1.2)
Collecting pip
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 2.4 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 23.1.2
Uninstalling pip-23.1.2:
Successfully uninstalled pip-23.1.2
Successfully installed pip-26.2.2
(.venv) PS E:\python-da> python -m pip install jupyterlab numpy pandas
Collecting jupyterlab
Downloading jupyterlab-4.6.4-py3-none-any.whl.metadata (16 kB)
Collecting numpy
Downloading numpy-2.4.6-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Collecting pandas
Downloading pandas-3.1.0-cp311-cp311-win_amd64.whl.metadata (18 kB)
Collecting async-lru>=1.0.0 (from jupyterlab)
Downloading async_lru-2.4.0-py3-none-any.whl.metadata (10 kB)
Collecting httpx<1,>=0.25.0 (from jupyterlab)
Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting ipykernel!=6.30.0,>=6.5.0 (from jupyterlab)
Downloading ipykernel-7.4.0-py3-none-any.whl.metadata (3.4 kB)
Collecting jinja2>=3.0.3 (from jupyterlab)
Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting jupyter-builder>=1.0.2 (from jupyterlab)
Downloading jupyter_builder-1.2.3-py3-none-any.whl.metadata (7.8 kB)
Collecting jupyter-core (from jupyterlab)
Downloading jupyter_core-5.9.1-py3-none-any.whl.metadata (1.5 kB)
Collecting jupyter-lsp>=2.0.0 (from jupyterlab)
Downloading jupyter_lsp-2.3.1-py3-none-any.whl.metadata (1.8 kB)
Collecting jupyter-server<3,>=2.19.0 (from jupyterlab)
Downloading jupyter_server-2.21.1-py3-none-any.whl.metadata (8.5 kB)
Collecting jupyterlab-server<3,>=2.28.0 (from jupyterlab)
Downloading jupyterlab_server-2.28.1-py3-none-any.whl.metadata (5.8 kB)
Collecting notebook-shim>=0.2 (from jupyterlab)
Downloading notebook_shim-0.2.4-py3-none-any.whl.metadata (4.0 kB)
Collecting packaging>=23.2 (from jupyterlab)
Downloading packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting tornado>=6.2.0 (from jupyterlab)
Downloading tornado-6.6-cp311-abi3-win_amd64.whl.metadata (2.7 kB)
Collecting traitlets (from jupyterlab)
Downloading traitlets-5.16.1-py3-none-any.whl.metadata (10 kB)
Collecting typing-extensions>=4.4.0 (from jupyterlab)
Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting anyio (from httpx<1,>=0.25.0->jupyterlab)
30°C
Partly sunny
Search
ENG
IN
99%
16:00
26-10-2026

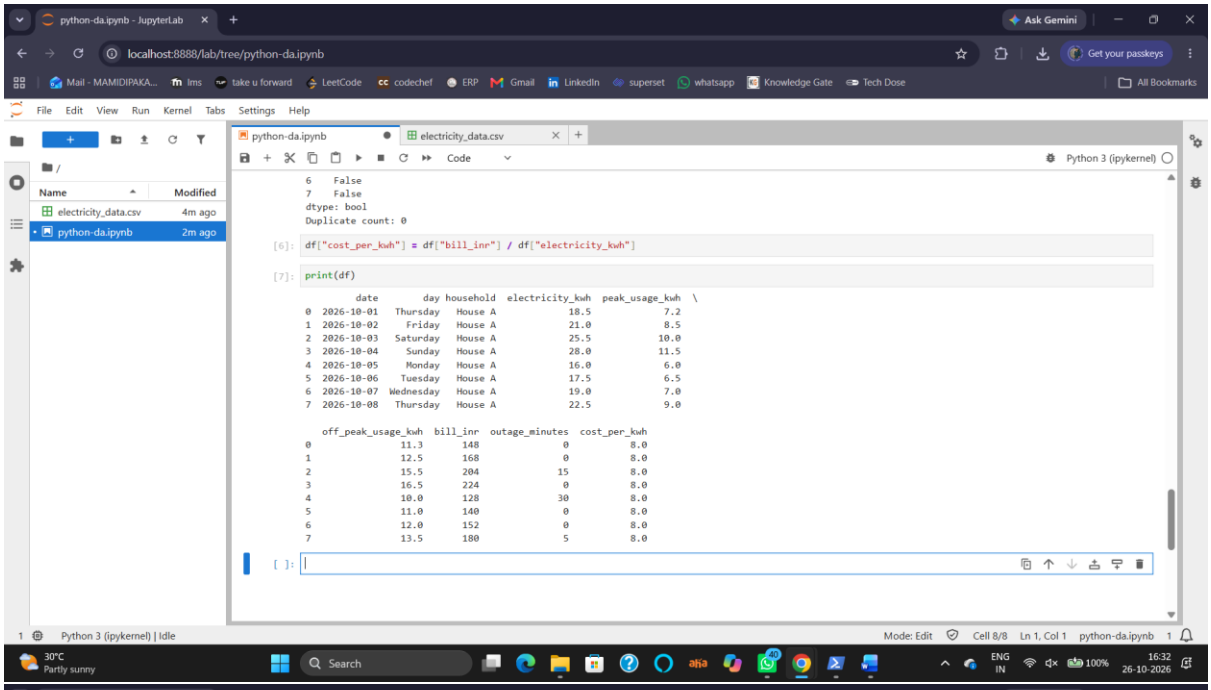

Windows PowerShell
Downloading rfc3987_syntax-1.1.0-py3-none-any.whl (8.0 kB)
Downloading lark-1.3.1-py3-none-any.whl (113 kB)
Downloading rpds_py-2026.9.1-cp311-cp311-win_amd64.whl (226 kB)
Downloading send2trash-2.1.0-py3-none-any.whl (17 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading stack_data-0.6.3-py3-none-any.whl (24 kB)
Downloading asttokens-3.0.2-py3-none-any.whl (28 kB)
Downloading executing-2.3.0-py3-none-any.whl (29 kB)
Downloading terminado-0.18.1-py3-none-any.whl (14 kB)
Downloading tinycss2-1.5.1-py3-none-any.whl (28 kB)
Downloading tornado-6.6-cp311-abi3-win_amd64.whl (530 kB)
---------- 530.9/530.9 kB 8.7 MB/s 0:00:00
Downloading traitlets-5.16.1-py3-none-any.whl (86 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading wcwidth-0.9.4-cp310-abi3-win_amd64.whl (597 kB)
---------- 597.8/597.8 kB 7.1 MB/s 0:00:00
Downloading webcolors-25.10.0-py3-none-any.whl (14 kB)
Downloading webencodings-0.6.1-py3-none-any.whl (8.7 kB)
Downloading websocket_client-1.9.2-py3-none-any.whl (95 kB)
Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-win_amd64.whl (25 kB)
Downloading cffi-2.1.1-cp311-cp311-win_amd64.whl (185 kB)
Downloading beautifulsoup4-4.15.0-py3-none-any.whl (109 kB)
Downloading soupsieve-2.10-py3-none-any.whl (38 kB)
Downloading defusedxml-0.7.1-py2.py3-none-any.whl (25 kB)
Downloading fqdn-1.6.0-py3-none-any.whl (12 kB)
Downloading isoduration-20.11.0-py3-none-any.whl (11 kB)
Downloading arrow-1.4.0-py3-none-any.whl (68 kB)
Downloading jupyterlab_pygments-0.3.0-py3-none-any.whl (15 kB)
Downloading pure_eval-0.2.4-py3-none-any.whl (11 kB)
Downloading pycparser-3.12-py3-none-any.whl (62 kB)
Downloading rfc3339_validator-0.1.4-py2.py3-none-any.whl (3.5 kB)
Downloading tzdata-2026.5-py2.py3-none-any.whl (347 kB)
Downloading uri_template-1.3.0-py3-none-any.whl (11 kB)
Installing collected packages: pure-eval, websocket-client, webencodings, webcolors, wcwidth, urllib3, uri-template, tzdata, typing-extensions, traitlets, tornado, soupsieve, six, send2trash, rpds-py, rfc3986-validator, pyzmq, pyyaml, pywinpty, python-json-logger, pygments, pycparser, psutil, prometheus-client, platformdirs, parso, pandocfilters, packaging, overrides, numpy, nest-asyncio2, mistune, MarkupSafe, lark, jupyterlab-pygments, jsonpointer, json5, idna, h11, fqdn, fastjsonschema, executing, defusedxml, debugpy, comm, colorama, charset_normalizer, certifi, babel, attrs, async-lru, asttokens, tinycss2, terminado, stack_data, rfc3987-syntax, rfc3339-validator, requests, referencing, python-dateutil, prompt_toolkit, matplotlib-inline, jupyter-core, jinja2, jedi, ipython-pygments-lexers, httpcore, cffi, bleach, beautifulsoup4, anyio, pandas, jupyter-server-terminals, jupyter-client, jupyter-builder, jsonschema-specifications, ipython, httpx, arrow, argon2-cffi-bindings, jsonschema, isoduration, ipykernel, argon2-cffi, nbformat, nbclient, jupyter-events, nbconvert, jupyter-server, notebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab
Successfully installed MarkupSafe-3.0.4 anyio-4.15.1 argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 arrow-1.4.0 asttokens-3.0.2 async-lru-2.4.0 attrs-26.1.0 babel-2.18.0 beautifulsoup4-4.15.0 bleach-6.4.0 certifi-2026.10.23 cffi-2.1.1 charset_normalizer-3.5.2 colorama-0.4.6 comm-0.2.3 debugpy-1.8.22 defusedxml-0.7.1 executing-2.3.0 fastjsonschema-2.22.2 fqdn-1.6.0 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.20 ipykernel-7.4.0 ipython-9.17.1 ipython-pygments-lexers-1.1.1 isoduration-20.11.0 jedi-0.20.1 jinja2-3.1.6 json5-0.17.3 jsonpointer-3.5 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 jupyter-builder-1.2.3 jupyter-client-8.10.0 jupyter-core-5.9.1 jupyter-events-0.12.1 jupyter-lsp-2.3.1 jupyter-server-2.21.1 jupyter-server-terminals-0.5.4 jupyterlab-4.6.4 jupyterlab-pygments-0.3.0 jupyterlab-server-2.28.1 lark-1.3.1 matplotlib-inline-0.2.2 mistune-3.4.0 nbclient-0.11.0 nbconvert-7.17.2 nbformat-5.11.1 nest-asyncio2-1.7.4 notebook-shim-0.2.4 numpy-2.4.6 overrides-7.7.0 packaging-26.3 pandas-3.1.0 pandocfilters-1.5.1 parso-0.8.7 platformdirs-4.13.5 prometheus-client-0.26.0 prompt_toolkit-3.0.53 psutil-7.2.2 pure-eval-0.2.4 pycparser-3.12 pygments-2.21.0 python-dateutil-3.0.0 python-json-logger-4.2.0 pywinpty-3.1.0 pyyaml-6.0.3 pyzmq-27.2.0 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.9.1 send2trash-2.1.0 six-1.17.0 soupsieve-2.10 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.6 traitlets-5.16.1 typing-extensions-4.16.0 tzdata-2026.5 uri-template-1.3.0 urllib3-2.8.0 wcwidth-0.9.4 webcolors-25.10.0 webencodings-0.6.1 websocket-client-1.9.2
(.venv) PS E:\python-da>
30°C
Partly sunny
Search
ENG
IN
99%
16:00
26-10-2026
Product Analytics Analysis
python-da.ipynb - JupyterLab
Ask Gemini
localhost:8888/lab/tree/python-da.ipynb
Get your passkeys
Mail - MAMIDIPAKA...
lms
take u forward
LeetCode
codechef
ERP
Gmail
LinkedIn
superset
whatsapp
Knowledge Gate
Tech Dose
All Bookmarks
File Edit View Run Kernel Tabs Settings Help
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb 10s ago
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
date day household electricity_kwh peak_usage_kwh \
0 2026-10-01 Thursday House A 18.5 7.2
1 2026-10-02 Friday House A 21.0 8.5
2 2026-10-03 Saturday House A 25.5 10.0
3 2026-10-04 Sunday House A 28.0 11.5
4 2026-10-05 Monday House A 16.0 6.0
5 2026-10-06 Tuesday House A 17.5 6.5
6 2026-10-07 Wednesday House A 19.0 7.0
7 2026-10-08 Thursday House A 22.5 9.0
off_peak_usage_kwh bill_inr outage_minutes
0 11.3 148 0
1 12.5 168 0
2 15.5 204 15
3 16.5 224 0
4 10.0 128 30
5 11.0 140 0
6 12.0 152 0
7 13.5 180 5
[2]:
daily_usage = np.array(df["electricity_kwh"])
daily_bill = np.array(df["bill_inr"])
daily_outage = np.array(df["outage_minutes"])
print("Daily usage:", daily_usage)
print("Daily bill:", daily_bill)
print("Daily outage:", daily_outage)
Daily usage: [18.5 21. 25.5 28. 16. 17.5 19. 22.5]
Daily bill: [148 168 204 224 128 140 152 180]
Daily outage: [0 0 15 0 30 0 0 5]
[ ]:
1 Python 3 (ipykernel) | Idle
Mode: Edit
Cell 3/3 Ln 1, Col 2 python-da.ipynb 1
30°C
Partly sunny
Search
ENG
IN
99%
16:25
26-10-2026

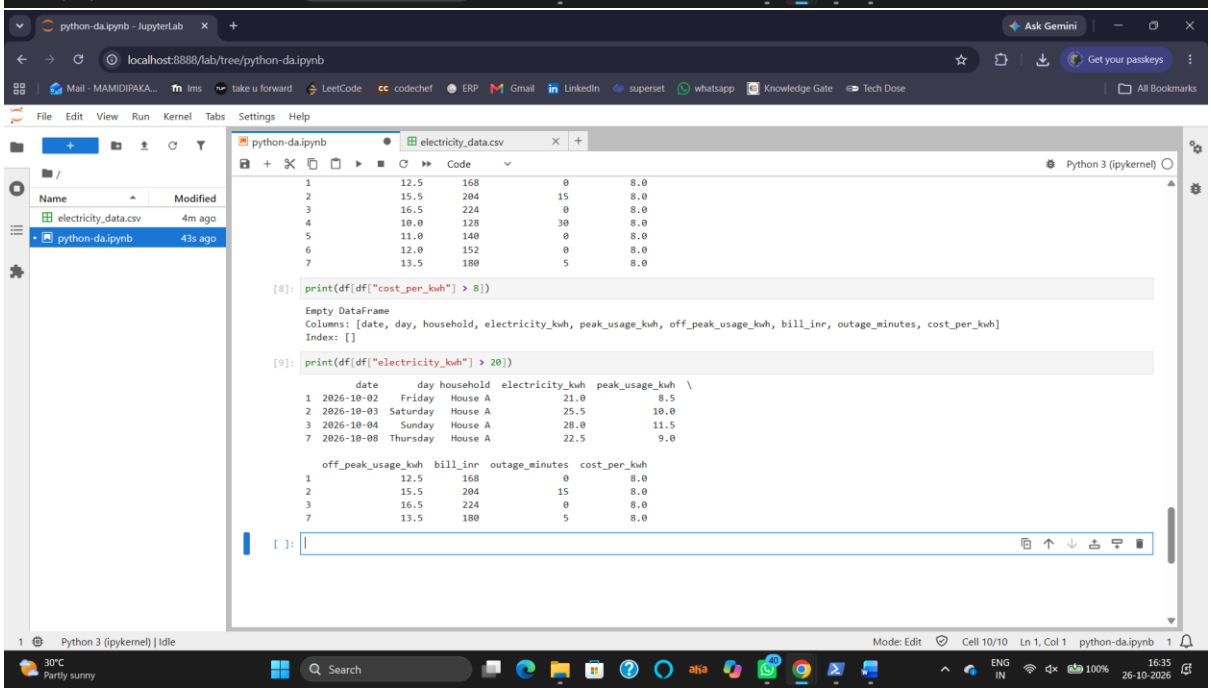

python-da.ipynb - JupyterLab
Ask Gemini
localhost:8888/lab/tree/python-da.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb now
[3]:
print("Total usage:", np.sum(daily_usage))
print("Minimum usage:", np.min(daily_usage))
print("Maximum usage:", np.max(daily_usage))
print("Mean usage:", np.mean(daily_usage))
Total usage: 168.0
Minimum usage: 16.0
Maximum usage: 28.0
Mean usage: 21.0
[4]:
print(df.head())
print(df.tail())
print("Shape:", df.shape)
print("Columns:", df.columns)
print("Column names:", df.columns.tolist())
df.info()
print(df.describe())
date day household electricity_kwh peak_usage_kwh \
0 2026-10-01 Thursday House A 18.5 7.2
1 2026-10-02 Friday House A 21.0 8.5
2 2026-10-03 Saturday House A 25.5 10.0
3 2026-10-04 Sunday House A 28.0 11.5
4 2026-10-05 Monday House A 16.0 6.0
off_peak_usage_kwh bill_inr outage_minutes
0 11.3 148 0
1 12.5 168 0
2 15.5 204 15
3 16.5 224 0
4 10.0 128 30
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 5/5
Ln 1, Col 1
python-da.ipynb
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26-10-2026

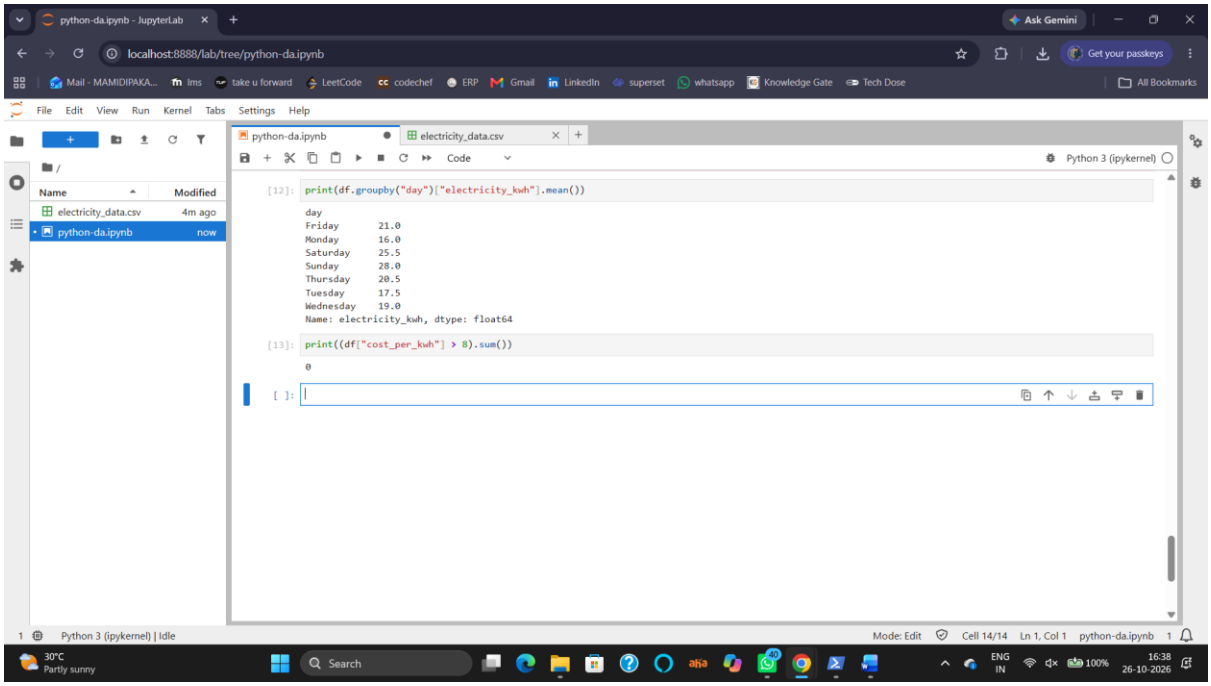

python-da.ipynb - JupyterLab
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File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb now
RangeIndex: 8 entries, 0 to 7
Data columns (total 8 columns):
Column Non-Null Count Dtype
0 date 8 non-null str
1 day 8 non-null str
2 household 8 non-null str
3 electricity_kwh 8 non-null float64
4 peak_usage_kwh 8 non-null float64
5 off_peak_usage_kwh 8 non-null float64
6 bill_inr 8 non-null int64
7 outage_minutes 8 non-null int64
dtypes: float64(3), int64(2), str(3)
memory usage: 644.0 bytes
electricity_kwh peak_usage_kwh off_peak_usage_kwh bill_inr \
count 8.000000 8.000000 8.000000 8.000000
mean 21.000000 8.212500 12.787500 168.000000
std 4.123106 1.890909 2.252895 32.984845
min 16.000000 6.000000 10.000000 128.000000
25% 18.250000 6.875000 11.225000 146.000000
50% 20.000000 7.850000 12.250000 160.000000
75% 23.250000 9.250000 14.000000 186.000000
max 28.000000 11.500000 16.500000 224.000000
outage_minutes
count 8.000000
mean 6.250000
std 10.938138
min 0.000000
25% 0.000000
50% 0.000000
75% 7.500000
max 30.000000
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 5/5
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python-da.ipynb
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26-10-2026


python-da.ipynb - JupyterLab
Ask Gemini
localhost:8888/lab/tree/python-da.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb now
[5]:
print(df.isnull())
print(df.isnull().sum())
print("Total missing values:", df.isnull().sum().sum())
print("Duplicate rows:")
print(df.duplicated())
print("Duplicate count:", df.duplicated().sum())
date day household electricity_kwh peak_usage_kwh \
0 False False False False False
1 False False False False False
2 False False False False False
3 False False False False False
4 False False False False False
5 False False False False False
6 False False False False False
7 False False False False False
off_peak_usage_kwh bill_inr outage_minutes
0 False False False
1 False False False
2 False False False
3 False False False
4 False False False
5 False False False
6 False False False
7 False False False
date 0
day 0
household 0
electricity_kwh 0
peak_usage_kwh 0
off_peak_usage_kwh 0
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 6/6
Ln 1, Col 1
python-da.ipynb
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Search
ENG
IN
16:29
26-10-2026

python-da.ipynb - JupyterLab
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localhost:8888/lab/tree/python-da.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb now
6 False False False False False
7 False False False False False
off_peak_usage_kwh bill_inr outage_minutes
0 False False False
1 False False False
2 False False False
3 False False False
4 False False False
5 False False False
6 False False False
7 False False False
date 0
day 0
household 0
electricity_kwh 0
peak_usage_kwh 0
off_peak_usage_kwh 0
bill_inr 0
outage_minutes 0
dtype: int64
Total missing values: 0
Duplicate rows:
0 False
1 False
2 False
3 False
4 False
5 False
6 False
7 False
dtype: bool
Duplicate count: 0
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 6/6
Ln 1, Col 1
python-da.ipynb
30°C
Partly sunny
Search
ENG
IN
16:29
26-10-2026


python-da.ipynb - JupyterLab
localhost:8888/lab/tree/python-da.ipynb
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb 2m ago
6 False
7 False
dtype: bool
Duplicate count: 0
[6]: df["cost_per_kwh"] = df["bill_inr"] / df["electricity_kwh"]
[7]: print(df)
date day household electricity_kwh peak_usage_kwh \
0 2026-10-01 Thursday House A 18.5 7.2
1 2026-10-02 Friday House A 21.0 8.5
2 2026-10-03 Saturday House A 25.5 10.0
3 2026-10-04 Sunday House A 28.0 11.5
4 2026-10-05 Monday House A 16.0 6.0
5 2026-10-06 Tuesday House A 17.5 6.5
6 2026-10-07 Wednesday House A 19.0 7.0
7 2026-10-08 Thursday House A 22.5 9.0
off_peak_usage_kwh bill_inr outage_minutes cost_per_kwh
0 11.3 148 0 8.0
1 12.5 168 0 8.0
2 15.5 204 15 8.0
3 16.5 224 0 8.0
4 10.0 128 30 8.0
5 11.0 140 0 8.0
6 12.0 152 0 8.0
7 13.5 180 5 8.0
[ ]:
Python 3 (ipykernel) | Idle
Mode: Edit Cell 8/8 Ln 1, Col 1 python-da.ipynb
30°C Partly sunny
Search
ENG IN
16:32 26-10-2026


python-da.ipynb - JupyterLab
localhost:8888/lab/tree/python-da.ipynb
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb 43s ago
1 12.5 168 0 8.0
2 15.5 204 15 8.0
3 16.5 224 0 8.0
4 10.0 128 30 8.0
5 11.0 140 0 8.0
6 12.0 152 0 8.0
7 13.5 180 5 8.0
[8]: print(df[df["cost_per_kwh"] > 8])
Empty DataFrame
Columns: [date, day, household, electricity_kwh, peak_usage_kwh, off_peak_usage_kwh, bill_inr, outage_minutes, cost_per_kwh]
Index: []
[9]: print(df[df["electricity_kwh"] > 20])
date day household electricity_kwh peak_usage_kwh \
1 2026-10-02 Friday House A 21.0 8.5
2 2026-10-03 Saturday House A 25.5 10.0
3 2026-10-04 Sunday House A 28.0 11.5
7 2026-10-08 Thursday House A 22.5 9.0
off_peak_usage_kwh bill_inr outage_minutes cost_per_kwh
1 12.5 168 0 8.0
2 15.5 204 15 8.0
3 16.5 224 0 8.0
7 13.5 180 5 8.0
[ ]:
Python 3 (ipykernel) | Idle
Mode: Edit Cell 10/10 Ln 1, Col 1 python-da.ipynb
30°C Partly sunny
Search
ENG IN
16:35 26-10-2026


python-da.ipynb - JupyterLab
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localhost:8888/lab/tree/python-da.ipynb
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python-da.ipynb
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
electricity_data.csv 4m ago
python-da.ipynb now
[12]: print(df.groupby("day")["electricity_kwh"].mean())
day
Friday 21.0
Monday 16.0
Saturday 25.5
Sunday 28.0
Thursday 20.5
Tuesday 17.5
Wednesday 19.0
Name: electricity_kwh, dtype: float64
[13]: print((df["cost_per_kwh"] > 8).sum())
0
[ ]:
1 Python 3 (ipykernel) | Idle
Mode: Edit
Cell 14/14
Ln 1, Col 1
python-da.ipynb
1
30°C
Partly sunny
Search
ENG
IN
100%
16:38
26-10-2026